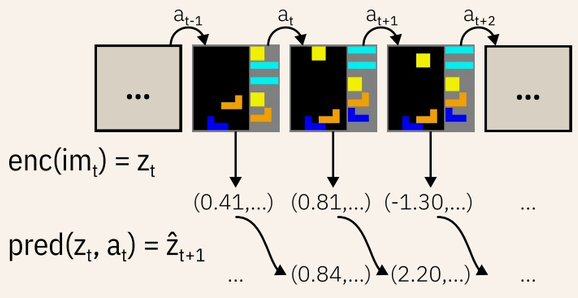
Learning to Play Tetris with World Models

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Background

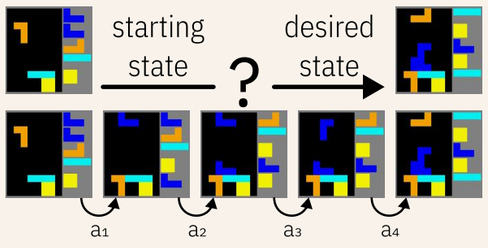
JEPA World Models

Jointly train encoder and predictor to learn the dynamics of an environment:



Latent Planning

Using the trained world model and a solver, plan an action sequence leading from starting to desired state:



Motivation

- State-of-the-art world models struggle to learn and to plan in dense 2D environments.
- Tetris is the perfect environment to stress-test this capability.
- Previous works only evaluate world models on planning from a starting to a desired state, but not higher-level goals, e.g. maximizing row clears in Tetris.

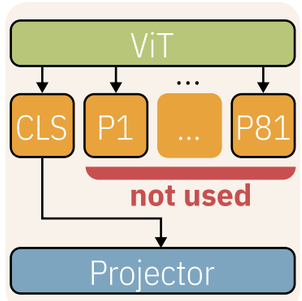
We contribute a simple architectural change for improved performance (patch projector), and a brand new, challenging world model environment (Tetris)

Architecture

Our key insight is that world model architectures are limited by the use of ViT patch tokens in their encoders.

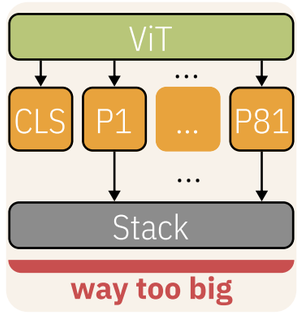
LeWM

Latent space **insufficiently expressive** due to missing patch tokens.



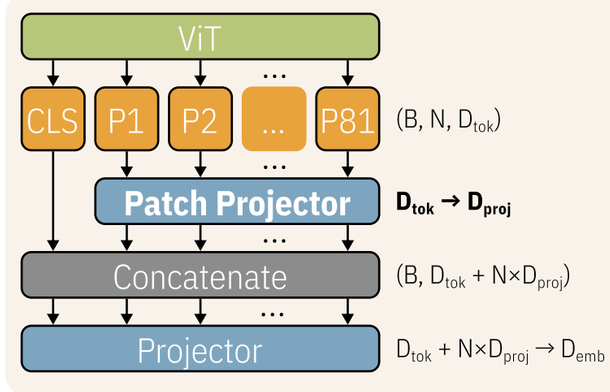
DINO-WM

Latent space **unworkably large** due to being just stacked patch tokens.



Ours

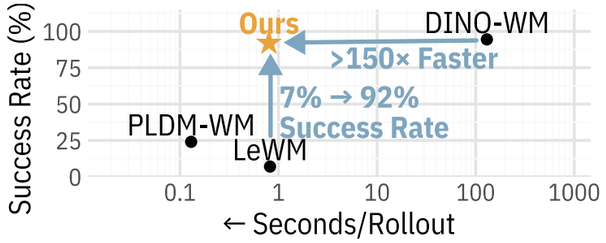
- We add a **patch projector**, a **single shared linear layer** for shrinking the dimension of patch tokens.
- The result is the best of both worlds: a latent space that is simultaneously expressive and compact.



Results

Arch.	Pred. Loss	Probe F1	Seconds /Rollout	Success Rate (%)
PLDM-WM	-	0.739	**0.13**	24.0
DINO-WM	0.054	**0.974**	129.88	**94.5**
LeWM	0.031	0.513	0.82	7.0
Ours	**0.021**	0.949	0.80	92.0

Latent Planning



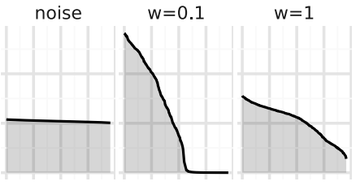
LeWM enhanced with a patch projector finds successful trajectories in a minimal amount of time.

Latent Probing

Probe F1 correlates significantly closer with actual downstream success rate compared to prediction loss i.e. the main training objective.

SIGReg Ablation

λ	Pred. Loss	Probe F1	Success Rate (%)
0.05	**0.0030**	0.670	39.0
0.1	0.0072	0.789	54.0
1.0	0.0304	**0.949**	**90.5**
1.25	0.0321	0.921	88.0
1.6	0.0377	0.911	85.0



For Tetris, tuning SIGReg weight to 1 maximizes probe F1 and latent diversity. But other envs become unstable.

Post-Training

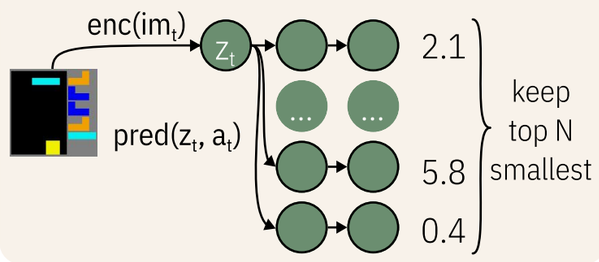
λ	Post Train	Pred. Loss	Probe F1	Success Rate (%)
0.1	no	**0.0072**	0.789	54.5
1	no	0.0304	**0.949**	90.5
1	yes	0.0213	**0.949**	**92.0**

Post-training alleviates prediction loss degradation under high SIGReg weight.

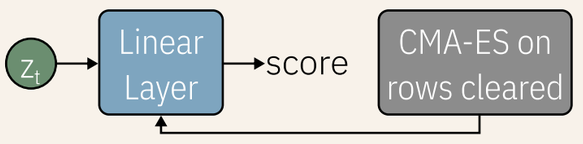
Playing Tetris

- To play Tetris, we use beam search minimizing the score resulting from mapping the latent state through a linear layer.
- To optimize the weights of the linear layer, we use CMA-ES.
- This simple setup is sufficient to fully solve Tetris with a reduced number of tetrominoes.

Beam Search over Actions



Learned Linear Layer Scorer



Tetrominoes			
Result	Plays forever	Plays forever	Median zero rows cleared, rarely one or two

Beyond Tetris

Train Dataset	Patch Proj.	Pred. Loss	Success Rate (%)
pusht	no	0.1531	2.30 ± 0.53
pusht	yes	**0.0059**	**3.80 ± 0.58**
reacher	no	0.0204	13.20 ± 1.21
reacher	yes	**0.0175**	**13.70 ± 1.13**
tworoom	no	0.0146	**37.70 ± 1.62**
tworoom	yes	**0.0051**	32.00 ± 1.41

Adding a patch projector results in a slight success rate improvement for some (pusht, reacher), but not all (tworoom) environments.